

ROSHAN DHARAN

Los Angeles, CA 90007 | (213) 930-3816 | rshashid@usc.edu | LinkedIn | GitHub | Portfolio

EDUCATION

University of Southern California

Master of Science in Computer Science

Aug 2025 – Present

Los Angeles, CA

REVA University

Bachelor of Technology in Computer Science and Engineering, GPA: 3.97/4

Jul 2019 – Jul 2023

Bengaluru, India

- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming (Python/Java/C++), Advanced DBMS, Operating Systems, Machine Learning, Artificial Intelligence, Cloud Computing, Big Data, Data Science with R

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, CCL), Java, C++, JavaScript, MATLAB

Backend / Agentic Tools: Spring Boot, FastAPI, LangChain, LangGraph, Java FX, CI/CD (GitHub Actions), Mockito, Cucumber

Big Data/Cloud: Databricks, Snowflake, GCP, AWS, OCI, Oracle Cerner Millennium EHR

ML/AI Tools: TensorFlow, Keras, PyTorch, XGBoost, ChromaDB, Claude/Anthropic API, OpenCV for multimodal predictive modeling, NLP

EXPERIENCE

Keck Medicine of USC

Software Engineer Intern, Clinical Applications

Nov 2025 – Present

Los Angeles, CA

- Spearheaded development of clinical applications integrated with Cerner Millennium EHR, implementing backend and UI improvements to support workflows for 50+ clinicians while ensuring HIPAA compliance.

Oracle

Associate Software Engineer, AMS Custom Reporting

Oct 2023 – Jul 2025

Bengaluru, India

- Led advanced analytics initiatives at Children's National Hospital across 17+ departments, driving 30–40% reduction in report runtimes and reclaiming 500+ staff hours/year by leveraging predictive models and automation.
- Managed a team of 6 to design MedeAnalytics SQL pipelines, processing 600,000+ patient records daily, performing data wrangling and ETL, and delivering financial metrics shaping \$200M+ revenue decisions.
- Created and transmitted 340B Extract to vendor using SQL, MoveIT, and Pharmacy Clinical Surveillance; streamlined workflow reclaimed 114 hours/year and cut pharmacy drug spend by \$12M+ annually.
- Developed CLABSI Audit system using Infusion Suite, Mobile Vitals Collection, and PowerChart, monitoring 500+ patients weekly, freeing nurses from 15+ hours/week of manual checks.

Oracle Cerner

Analytics Intern, International Consulting PMO

Jan 2023 – Jul 2023

Bengaluru, India

- Migrated 5 enterprise-level Power BI reports from on-premise and SharePoint platforms to Snowflake Cloud, centralizing fragmented data by rebuilding complex KPIs with DAX. Automated offline-to-online data entry via Python multithreading + Selenium, cutting manual processing by 44 hours per 1,000 rows.

PROJECTS

Hospital AI Command Center — Python, LangGraph, XGBoost, Claude, FastAPI

2026

- Built an 8-node LangGraph pipeline that fires on every HL7 ADT event, running XGBoost risk scoring across 5 clinical dimensions with SHAP explainability.
- Retrieved CMS/HEDIS guidance via a ChromaDB knowledge base and synthesized Claude-generated intervention plans, pushed via SSE to a live dashboard with full action-accountability logging; deployed on Railway.

RAG-Powered Codebase Q&A Agent — Python, LangChain, Claude, FastAPI

2026

- Built a citation-grounded RAG pipeline over any public GitHub repo using language-aware chunking and MMR retrieval to reduce redundant context.
- Cached local embeddings for cheap re-ingestion and shipped a retrieval evaluation harness, CI, and a Docker Compose deployment with a Streamlit chat UI.

ICU Outcome Prediction — Sklearn, PyTorch, Streamlit

Jan 2025 – May 2025

- Built a multimodal ICU outcome predictor combining LSTM vital sign modeling with BioClinicalBERT embeddings, improving AUROC by 6–8% over NLP baselines.
- Designed and implemented end-to-end ML workflows, including TF-IDF feature extraction from 5,000+ clinical note features with logistic regression, deployed as a REST API service with CI/CD integration for reproducibility.

HONORS & PUBLICATIONS

Oracle Pinnacle Award — Invented memory allocation method adopted org-wide.

Jun 2025

"AI-Powered Battery Management System," peer-reviewed publication on EV battery SoC/SoH prediction —

2023

International Journal of Computational Learning & Intelligence